Fall 2024 Junior Topology Talk — Exotic 7-Spheres

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These are notes for a talk in the Junior Topology seminar. Much of the content is synthesized from Milnor's original paper ("On Manifolds Homeomorphic to the 7-sphere"), a UChicago REU paper by Rachel McEnroe, and Abhishek's notes on the same subject.

1 Sphere Bundles over Spheres

Our main guiding examples are the *Hopf fibrations*.

Example 1.

Hopf Fibration *The Hopf fibration is an S^1 bundle over S^2 with total space S^3. The projection map is easily described if we interpret S^3 as the unit sphere in $\mathbb{C}^2$ and S^2 as $\mathbb{CP}^1$ — we have*

$$\begin{tikzcd} U(1) = S^1 \arrow[r, hook] & S^3 \subset \mathbb{C}^2 \arrow[d, "(z,w)\mapsto[z,w]"] \\ & S^2 = \mathbb{CP}^1 \end{tikzcd}$$

Quaternionic Hopf Fibration *There is a natural quaternionic analogue for the unit sphere in $\mathbb{H}^2$ and $S^4 = \mathbb{HP}^1$*

$$\begin{tikzcd} SU(2) = S^3 \arrow[r, hook] & S^7 \subset \mathbb{H}^2 \arrow[d, "(z,w)\mapsto[z,w]"] \\ & S^4 = \mathbb{HP}^1 \end{tikzcd}$$

We actually have a fairly comprehensive classification of S^n-bundles over spheres for $n = 1, 2, 3$.[1]

Fact. In these dimensions, it turns out that every S^n-bundle over S^m arises as the unit sphere subbundle of a rank n vector bundle over S^m, where $m < n$.

As such, it suffices to attempt to understand vector bundles over spheres.

1.1 Clutching Construction

Theorem 2. *For any n, vector bundles of rank n over S^m are classified by*

$$\pi_{m-1}\,\mathrm{GL}_n(\mathbb{R}) \cong \pi_{m-1}\,\mathrm{O}(n).$$

The idea behind this theorem is not terribly complicated. Fundamentally, vector bundles are built by gluing together trivial bundles over charts on the base space. Every sphere S^m can be covered by two charts: the

[1] It is a subtle point that this does not hold for $n \geq 4$. The failure of the high-dimensional Smale conjecture leads to both "exotic diffeomorphisms" of spheres and corresponding "exotic" sphere bundles which do not arise as the sphere subbundle of a vector bundle.

complement U_N, U_S of the north and south pole, respectively. The data of this gluing is in the form of the transition map $g : U_N \cap U_S \to GL_n(\mathbb{R})$. Note that $U_N \cap U_S$ is homotopy equivalent to the equator of S^m, namely the sphere of dimension one less S^{m-1}. Homotopic gluing maps yield the same vector bundle, and so vector bundles of rank n over S^m are classified by homotopy classes of maps $[S^{m-1}, \mathrm{GL}_n(\mathbb{R})]$, or $\pi_{m-1}\,\mathrm{GL}_n(\mathbb{R})$. Since $\mathrm{GL}_n(\mathbb{R})$ deformation retracts onto its maximal compact subgroup $O(n)$, the theorem holds.

Taking the unit sphere subbundle corresponds to reducing the structure group to $\mathrm{SO}(m)$, and so

Theorem 3. *For $n = 1, 2, 3$ and $m < n$, isomorphism classes of S^n bundles over S^m are in bijection with $\pi_{m-1}(\mathrm{SO}(n+1))$.*

Example 4.

Hopf Fibration *We can build the Hopf fibration using this clutching construction. Consider the maps $\varphi_N : U_N \to \mathbb{R}^2 = \mathbb{C}$ and $\varphi_S : U_S \to \mathbb{R}^2 = \mathbb{C}$ given by $[z; 1] \mapsto z$ and $[1; w] \mapsto w$, where as before U_N, U_S are the complements of the south and north pole, respectively. Note that the transition map*

$$\varphi_S \circ \varphi_N^{-1} : \mathbb{C} \setminus \{0\} \to \mathbb{C} \setminus \{0\}$$

is

$$z \mapsto \frac{1}{z}.$$

Now, we can think about how the Hopf fibration is actually glued together. First, note that the Hopf fibration is naturally the unit sphere subbundle of the canonical line bundle L over $\mathbb{C}\mathrm{P}^1$. We first see the gluing map for $\pi : L \to \mathbb{C}\mathrm{P}^1$, then normalize to find the gluing map of the Hopf fibration. Consider the following two local trivializations:

$$\begin{aligned} \rho_N : \underbrace{\pi^{-1}(U_N)}_{\subset \mathbb{C}^2 \times \mathbb{C}\mathrm{P}^1} &\to \varphi_N(U_N) \times \mathbb{C} & \qquad \rho_S : \pi^{-1}(U_S) &\to \varphi_S(U_S) \times \mathbb{C} \\ ((x, y), [z; 1]) &\mapsto (z, y) & ((x, y), [1; w]) &\mapsto (w, x). \end{aligned}$$

We see that

$$\begin{aligned} \rho_S \circ \rho_N^{-1} : \varphi_N(U_N \cap U_S) \times \mathbb{C} &\to \varphi_S(U_N \cap U_S) \times \mathbb{C}) \\ (z, x) &\mapsto \left(\frac{1}{z}, xz\right). \end{aligned}$$

We can restrict the fiber-wise maps to act on the unit complex numbers by normalizing:

$$\begin{aligned} \rho_S \circ \rho_N^{-1} : \varphi_N(U_N \cap U_S) \times \mathbb{C} &\to \varphi_S(U_N \cap U_S) \times \mathbb{C}) \\ (z, x) &\mapsto \left(\frac{1}{z}, \frac{xz}{\|z\|}\right). \end{aligned}$$

Quaternionic Hopf Fibration *The above analysis works exactly the same, replacing $\mathbb{C}$ with $\mathbb{H}$.*

2 A Family of Topological 7-Spheres

We will be especially interested in a family of bundles resembling the quaternionic Hopf fibration mentioned at the beginning. By the clutching construction, S^3 bundles over S^4 are in bijection with $\pi_3(SO(4))$. Let's compute this group!

The group $\mathrm{Spin}(n)$ is defined to be the double cover of $\mathrm{SO}(n)$. Relevant to us is that there is an exceptional isomorphism $\mathrm{Spin}(4) \cong \mathrm{SU}(2) \times \mathrm{SU}(2)$, and since $\mathrm{SU}(2)$ can be identified with the unit quaternions, $\mathrm{Spin}(4)$ is diffeomorphic to $S^3 \times S^3$. In particular, $S^3 \times S^3$ double covers SO(4). Explicitly, the map is given by

$$\begin{aligned} \Psi : S^3 \times S^3 &\to SO(4) \\ (z, w) &\mapsto \psi_{z,w} = (x \mapsto zxw^{-1}) \end{aligned}$$

A useful result from algebraic topology concludes the computation:

Theorem 5. *If $F : X \to Y$ is a covering map, then $\pi_n(X) \cong \pi_n(Y)$ for all $n \geq 2$.*

Proof. Since $\pi_1(S^n) = 1$ for all $n \geq 2$, the result follows from the homotopy lifting property. □

We conclude that $\pi_3(SO(4)) \cong \pi_3(S^3 \times S^3) \cong \mathbb{Z} \times \mathbb{Z}$.

So, we find that two integers, say h and j, determine S^3 bundles over S^4 — call the total space of these bundles $M_{h,j}$, and the bundle maps $\sigma_{h,j}$. Also denote the associated vector bundles as $\xi_{h,j}$. It's actually possible to compute the gluing maps for $M_{h,j}$ explicitly. Here's the plan: using the double cover map ψ, map points in S^3 to automorphisms of S^3. Then, thinking of $\mathbb{H} \setminus \{0\} \cong S^4 \setminus \{N, S\}$, extend this transition map to all of $\mathbb{H} \setminus \{0\}$ by normalizing and reflecting across the S^3 equator. One way to do this is as follows: consider the family of maps

$$\begin{aligned} \widetilde{f}_{h,j} : S^3 &\to S^3 \times S^3 \\ u &\mapsto (u^h, u^{-j}) \end{aligned}$$

Then the composition

$$\begin{aligned} f_{h,j} = \psi \circ \widetilde{f}_{h,j} : S^3 \to S^3 \times S^3 &\to \mathrm{SO}(4) \\ u &\mapsto (x \mapsto u^h x u^{-j}) \end{aligned}$$

is an isometry of S^3. Normalizing, we can extend this to a map

$$\begin{aligned} F_{i,j} : \mathbb{H} \setminus \{0\} &\to \mathrm{SO}(4) \\ z &\mapsto \left(x \mapsto \frac{z^h x z^j}{\|z\|^{h+j}} \right) \end{aligned}$$

Finally, we couple this with the standard transition maps between the charts U_N, U_S to get our gluing map:

$$\begin{aligned} \underbrace{\varphi(U_N \cap U_S)}_{\cong \mathbb{H} \setminus \{0\}} \times S^3 &\to \varphi(U_N \cap U_S) \times S^3 \\ (z, x) &\mapsto \left(\frac{1}{z}, F_{i,j}(z)(x) = \frac{z^h x z^j}{\|z\|^{h+j}} \right) \end{aligned}$$

By construction, these maps are not homotopic, since if they were, the homotopy would lift to a homotopy from S^3 to $S^3 \times S^3$, while $\widetilde{f}_{h,j}$ correspond to different elements of $\pi_3(S^3 \times S^3)$.

We are interested specifically in the case where $h + j = 1$. We will show that this (infinite) family is pairwise homeomorphic to S^7 but which is not diffeomorphic to S^7 if $(h - j)^2 \neq 1 \pmod 7$.

2.1 $M_{h,j}$ is homeomorphic to S^7 if $h + j = 1$

Theorem 6 (Reeb's Theorem)**.** *If M is a compact smooth n-manifold and $f : M \to \mathbb{R}$ is a Morse function with exactly two critical points, then M is homeomorphic to S^n.*

Proof. If f is as in the theorem statement, it must be that one of the critical points p is a minimum while the other q is a maximum. Thus, M is the union of an n-cell and a 0-cell, hence is homeomorphic to S^n. □

The main problem we face is coming up with a Morse function which satisfies the criterion of Reeb's theorem. Milnor gives us such a function: on the local trivializations ρ_N, ρ_S (as described earlier), define the function g such that

$$\begin{aligned} g_N : \varphi_N(U_N) \times S^3 &\to \mathbb{R} \\ (z, x) &\mapsto \frac{\mathrm{Re}(x)}{\sqrt{1 + \|z\|^2}} \\ g_S : \varphi_S(U_S) \times S^3 &\to \mathbb{R} \\ (w, y) &\mapsto \frac{\mathrm{Re}(wy^{-1})}{\sqrt{1 + \|wy^{-1}\|^2}} \end{aligned}$$

Although we don't have time to go through the computation right now, it turns out that g agrees on $\varphi_N(U_N \cap U_S)$ and $\varphi_S(U_N \cap U_S)$ if and only if $h + j = -1$. Moreover, one can check that g_N has critical points at $(0, 1)$ and $(0, -1)$, whereas g_S has no critical points. By Reeb's theorem, $M_{h,j}$ is homeomorphic to S^7 when $h + j = -1$. However, it also turns out that $M_{h,j} \cong M_{-h,-j}$, hence this holds when $h + j = 1$ as well.

3 Characteristic Classes

Characteristic classes are invariants of vector bundles, and they can be thought of as measuring "twistedness" and as obstructing the existence of certain kinds of sections. Here, we define the characteristic classes we need to argue that some of the $M_{h,j}$ are not diffeomorphic to S^7. We will need the *Euler class*, top and bottom *Chern class*, and the first *Pontryagin class*.

Definition 7. Consider some real orientable fiber bundle $\xi : E \to X^m$ of rank n over a compact orientable m-manifold X. If we consider the self-intersection of the zero section (so with a transversal perturbation of the zero section), we get a closed submanifold N of E of dimension $m - n$. Define the *Euler class* $e(\xi) \in H^n(X; \mathbb{Z})$ to be the Poincaré dual of $[N] \in H_{m-n}(X; \mathbb{Z})$.

Example 8. *(1) If $Y \subset X$ is a compact submanifold, then the Euler class of the normal bundle of Y in X is naturally identified with the self-intersection of Y in X.*

(2) The Euler class of the standard Hopf fibration is $-\alpha$, where α is a generator for $H^4(S^4; \mathbb{Z})$.

3.1 Chern classes

Now, consider some rank $2n$ complex vector bundle $\xi : E \to X^m$.

Definition 9. The *top Chern class* $c_n(\xi) \in H^{2n}(M; \mathbb{Z})$ is the Euler class $e(\xi_{\mathbb{R}}) \in H^{2n}(M; \mathbb{Z})$ of the underlying real bundle.

Recall, the *determinant bundle* of a vector bundle is defined to be its top exterior power $\wedge^n \xi$. Explicitly, its fibers are the (complex or real) lines spanned by $e_1 \wedge \cdots \wedge e_n$ where $\{e_i\}_{i=1}^n$ is a basis for the given fiber. Projection of course sends the same fibers to the same points. For complex bundles, $\wedge^n \xi$ will be a rank 1 complex bundle.

Definition 10. The *first Chern class* $c_1(\xi)$ is the Euler class $e((\wedge^n \xi)_{\mathbb{R}})$ of the underlying real bundle of the determinant bundle.

The intermediate Chern classes are a bit harder to define, and so we'll leave that to you if you're interested. Now, consider some real rank n vector bundle $\xi : E \to X$.

Definition 11. The *Pontryagin classes* $p_k(\xi) \in H^{4k}(X; \mathbb{Z})$ are defined to be $(-1)^k c_{2k}(\xi)$, where c_{2k} is the $2k$-th Chern class of the complex vector bundle $\xi \otimes \mathbb{C}$.

The (blank) class of a manifold M is defined to be the (blank) class of its tangent bundle. In particular, Chern classes ought to be defined for complex manifolds, or on complexified tangent bundles. There is a major result about Pontryagin classes which will be necessary for us to reach the desired conclusion:

Theorem 12 (Hirzebruch Signature Theorem)**.** *Let M be a closed orientable smooth manifold of dimension 8, with signature $\tau(M)$. Then*

$$\tau(M) = \frac{1}{45}(7p_2(M) - p_1^2(M)).$$

4 Family is not diffeomorphic to S^7

Our main objective is to construct a set-up to which we can apply this powerful tool. To do so, we consider the B^8 bundle $\mathfrak{b}_{h,j} : B_{h,j} \to S^4$ whose boundary is $M_{h,j}$ (taking fibers have norm ≤ 1 in $\mathbb{H}$). If $M_{h,j}$ is diffeomorphic to S^7, the boundary of $B_{h,j}$ is a sphere, and so we can glue an 8-ball to $B_{h,j}$ to create a new closed, smooth 8-manifold $C_{h,j}$. We will apply the Hirzebruch Signature Theorem to $C_{h,j}$ to contradict the assumption that $M_{h,j}$ is diffeomorphic to S^7.

We recall a few important results from the theory of universal bundles:

Theorem 13. $\pi_3(\mathrm{SO}(4)) \cong \pi_4(\mathrm{BSO}(4))$

This follows from the long exact sequence of homotopy groups.

Theorem 14. *Every rank 4 real vector bundle over S^4 is the pullback of the universal bundle* $\mathrm{ESO}(4)$ *over* $\mathrm{BSO}(4)$ *with respect to some continuous map (unique up to homotopy) from S^4 to* $\mathrm{BSO}(4)$*.*

Thus, we find a one-to-one correspondence between homotopy classes of maps from S^4 to $\mathrm{BSO}(4)$ and rank 4 real vector bundles over S^4.

$$\begin{array}{ccc} \pi^*(\mathrm{ESO}(4)) = \xi_{h,j} & \longrightarrow & \mathrm{ESO}(4) \\ \big\downarrow & & \big\downarrow \pi \\ S^4 & \xrightarrow[(h,j)\in\pi_4(\mathrm{BSO}(4))]{} & \mathrm{BSO}(4) \end{array}$$

4.1 Computing $p_1(\xi_{h,j})$

Lemma 15. *Characteristic classes commute with pullback. For a vector bundle $\xi : E \to B$ and a continuous map $f : B' \to B$, the pullback $f^*(\kappa(\xi))$ of the characteristic class is equal to the characteristic class of the pullback bundle $\kappa(f^*(\xi))$.*

Claim 16. *We claim that for any $\nu \in H^4(S^4;\mathbb{Z}) \cong \mathbb{Z}$ which is the pullback along some continuous map $h : S^4 \to$* $\mathrm{BSO}(4)$*, ν is linear in h and j. That is $\nu = (a \cdot h + b \cdot j) \cdot \alpha$, where α is a generator of $H^4(S^4;\mathbb{Z})$. In particular, this applies to any characteristic class of the universal bundle of* $\mathrm{BSO}(4)$*.*

Proof. Impressionistic version: For any $\nu \in H^4(BSO(4);\mathbb{Z})$, there is a natural homomorphism

$$\begin{aligned} \Theta_\nu : \pi_4(\mathrm{BSO}(4)) &\to H^4(S^4;\mathbb{Z}) \\ [h] &\mapsto h^*(\nu). \end{aligned}$$

Since the sum $[f] + [g] \in \pi_4(\mathrm{BSO}(4))$ is given by the homotopy class of the map

$$f + g : S^4 \to S^4 \vee S^4 \xrightarrow{f \vee g} \mathrm{BSO}(4),$$

where the first function is the pinch map, it can be shown that $\Theta_\nu([f] + [g]) = \Theta_\nu([f + g]) = \Theta_\nu([f]) + \Theta_\nu([g])$.

Moreover, we have the following map:

$$H^4(S^4) \times H^4(S^4) \xrightarrow{\cong} H^4(S^4 \vee S^4) \xrightarrow{\text{pinch}^*} H^4(S^4)$$
$$(\alpha, \beta) \mapsto \text{pinch}^*(\alpha) + \text{pinch}^*(\beta) \mapsto \alpha + \beta,$$

where the pinch map takes S^4 to $S^4 \vee S^4$ by "pinching" its equator to a point. This is enough to show the result.

Geometrically: if we have clutching functions $f_1, f_2 : S^3 \to \text{SO}(4)$, then we have fiber bundles E_1, E_2 over S^3. Multiplication $[f_1] \cdot [f_2]$ in $\pi_3(\text{SO}(4))$ is just the wedge $f_1 \vee f_2$, and this yields a clutching function on $S^3 \vee S^3$ and hence a fiber bundle E_3 on $S^4 \vee S^4$ (the suspension of $S^3 \vee S^3$ is two copies of S^4 glued along an interval, which is homotopy equivalent to $S^4 \vee S^4$). One can show that $c(E_3) = c(E_1) \oplus c(E_2)$.

The pinch map $S^4 \to S^4 \vee S^4$ induces the map $H^4(S^4 \vee S^4) \to H^4(S^4)$ given by taking $a \oplus b$ to $a + b$, and so brings $c(E_3)$ to $c(E_1) + c(E_2)$. Thus, $(h, j) + (h', j') = (h + h', j + j')$ in $\pi_3(SO(4))$ corresponds to the wedge of $f_{h,j}$ and $f_{h',j'}$, hence $p_1(\xi_{h,j})$ depends linearly on h and j. ☐

If we choose ν to be the Euler or Pontryagin class of the universal bundle of BSO(4), then it follows that their pullbacks, the classes of $\xi_{h,j}$, are linear in h and j.

Recall, there is an orientation-reversing isomorphism $\xi_{h,j} \cong \xi_{-h,-j}$. This implies that $e(\xi_{h,j}) = -e(\xi_{-j,-h})$. If we set $(h, j) = (0, 1)$, yielding the standard quaternionic Hopf fibration, then we see that there is an "inside out" Hopf fibration where $(h, j) = (-1, 0)$. So,

$$x \cdot h + y \cdot j = -(x \cdot (-j) + y \cdot (-h)).$$

The Euler class of the standard Hopf fibration is $-\alpha$, so

$$x \cdot 0 + y \cdot 1 = -1.$$

Combining these equations gives $x = y = -1$. So, for some generator α, we have that

$$e(\xi_{h,j}) = -(h + l)\alpha.$$

If we choose ν to be the Pontryagin class, we have that

$$p_1(\xi_{h,j}) = a \cdot h + b \cdot j.$$

Since Pontryagin classes are independent of the orientation of the bundle,

$$ah + bj = a(-j) + b(-h),$$

and so

$$p_1(\xi_{h,l}) = a(h - l)\alpha.$$

4.2 Computing $p_1(C_{h,j})$

The tangent bundle of $B_{h,j}$ naturally splits as the bundle of vectors tangent to each fiber and whose vectors are normal to each fiber, and using the formula for the Chern class of a Whitney sum (along with the fact that $p_1(TS^4) = 0$), we find that

$$p_1(B_{h,j}) = \mathfrak{b}_{h,j}^*(a(h - j)\alpha) = a(h - j)\beta$$

where $\beta = \mathfrak{b}_{h,j}^*(\alpha)$ is the generator of $H^4(B_{h,j})$. With some work, one can find that $a = \pm 2$.

Explicitly, we find that $TB_{h,j} \cong \mathfrak{b}^*(\xi_{h,j}) \oplus \mathfrak{b}^*(TS^4)$. If we denote ϵ^n as the trivial real vector bundle of rank n

over S^4, we find that since $TS^4 \oplus \epsilon^1 \cong \epsilon^5$,

$$\begin{aligned} TB_{h,j} \oplus \epsilon^1 &\cong \mathfrak{b}^*(\xi_{h,j}) \oplus \mathfrak{b}^*(TS^4) \oplus \epsilon^1 \\ &\cong \mathfrak{b}^*(\xi_{h,j}) \oplus \mathfrak{b}^*(TS^4 \oplus \epsilon^1) \\ &\cong \mathfrak{b}^*(\xi_{h,j}) \oplus \mathfrak{b}^*(\epsilon^5) \\ &\cong \mathfrak{b}^*(\xi_{h,j}) \oplus \epsilon^5. \end{aligned}$$

Applying the following lemma allows us to compute $p_1(TB_{h,j})$:

Lemma 17. *For any characteristic class κ of a real vector bundle ξ,*

$$\kappa(\xi) = \kappa(\xi \oplus \epsilon^n),$$

where ϵ^n is the trivial rank n real vector bundle over the base. Generalizing to complex vector bundles allows κ to be the Chern class as well.

So, $p_1(TB_{h,j}) = \mathfrak{b}^*(p_1(\xi_{h,j}))$.

The injection $\iota : B_{h,j} \to C_{h,j}$ induces an isomorphism on 4th cohomology and

$$\iota^*(TC_{h,j}) \cong TB_{h,j},$$

and so we find that $p_1(C_{h,j}) = 2(h-j)\beta$ where $\beta = \mathfrak{b}^*(\alpha)$.

4.3 Apply Hirzebruch

Now, since $H^4(C_{h,j}) = \mathbb{Z}$ (gluing on an 8-ball doesn't affect 4th cohomology), the signature is ± 1 since it comes from a 1×1-matrix. Applying the Hirzebruch Signature Theorem, we have that

$$\pm 1 = \frac{1}{45}(7p_2(C_{h,j}) - (\pm 2(h-j))^2).$$

Taking this equation mod 7, we can remove the reliance on p_2, and we find that

$$3 = \pm 4(h-j)^2 \pmod 7,$$

hence

$$(h-j)^2 = 1 \pmod 7.$$

This is a necessary condition for $C_{h,j}$ to have a smooth structure. Since both the 8-ball and $B_{h,j}$ have smooth structures, the contradiction must come from that fact that $\partial B_{h,j} = M_{h,j}$ can be glued via a diffeomorphism. Therefore, $M_{h,j}$ is not diffeomorphic to S^7 if $(h-j)^2 \neq 1 \pmod 7$. So, $M_{0,1}$ and $M_{1,0}$, the standard quaternionic Hopf fibrations *are* diffeomorphic to S^7, but $M_{3,-2}$ for example is not.

In summary, we have constructed a family of manifolds which are homeomorphic but not diffeomorphic to S^7.